PHY-112 | Principles of Physics-2

Akiful Islam (AZW)
Spring 2025 | Class #11

Department of Mathematics & Natural Sciences

BRAC UNIVERSITY
Inspiring Excellence

Enter Electric Current

BRAC UNIVERSITY
Inspiring Excellence

Electric Current: A Controlled Motion of Charges

Does it actually flow in and out of battery?

Define **current** as the flow of charge carriers within a conductor.

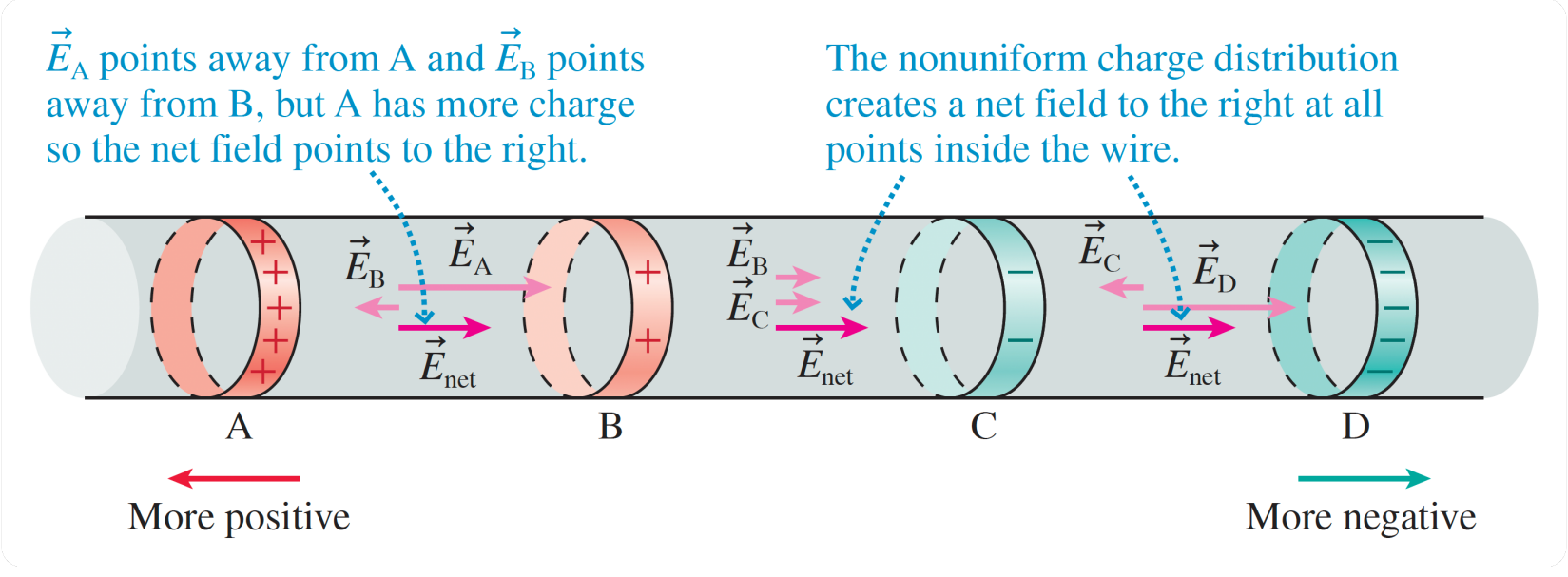
$$I = \frac{\Delta Q}{\Delta t}$$

- measured in [A] ampere
- is a scalar
- does not change its motion path *yet*
- heats up the conductor

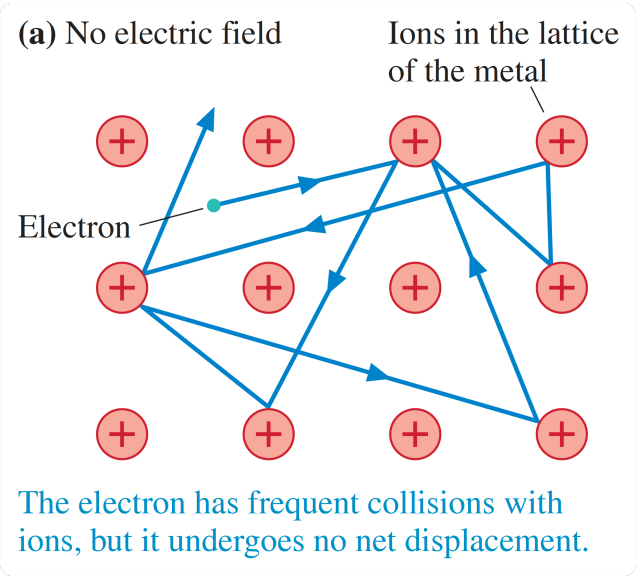
Establishing an Electric Field in a Wire

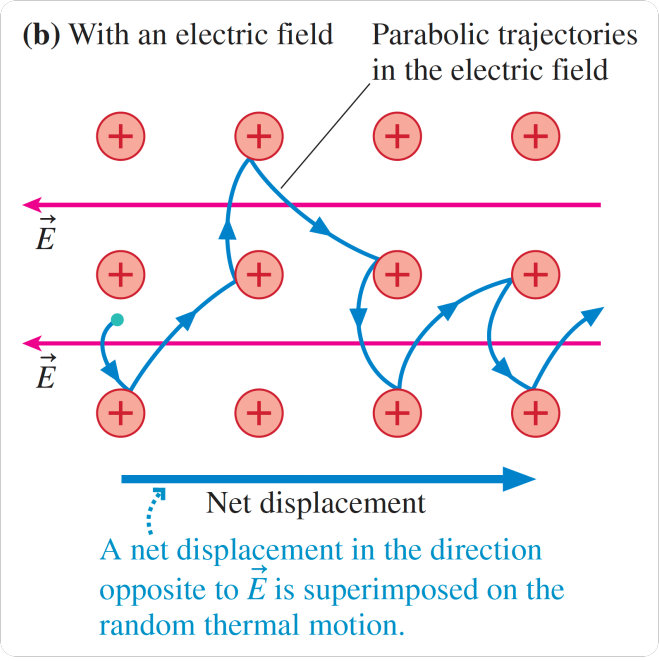
We need to be PERSISTENT with this push!

A *nonuniform* distribution of surface charges along a wire creates a net electric field $\vec{E}_{\text{net}}$ inside the wire that points from the more positive end of the wire toward the more negative end of the wire. This is the internal electric field that pushes the electron current through the wire.



Theory of Metallic (Microscopic) Conduction
On an atomic scale!





The steady electric force causes the electrons to move along *parabolic trajectories* between collisions.

We need to supply some Power!

BRAC UNIVERSITY
Inspiring Excellence

Sources of Electric Potential

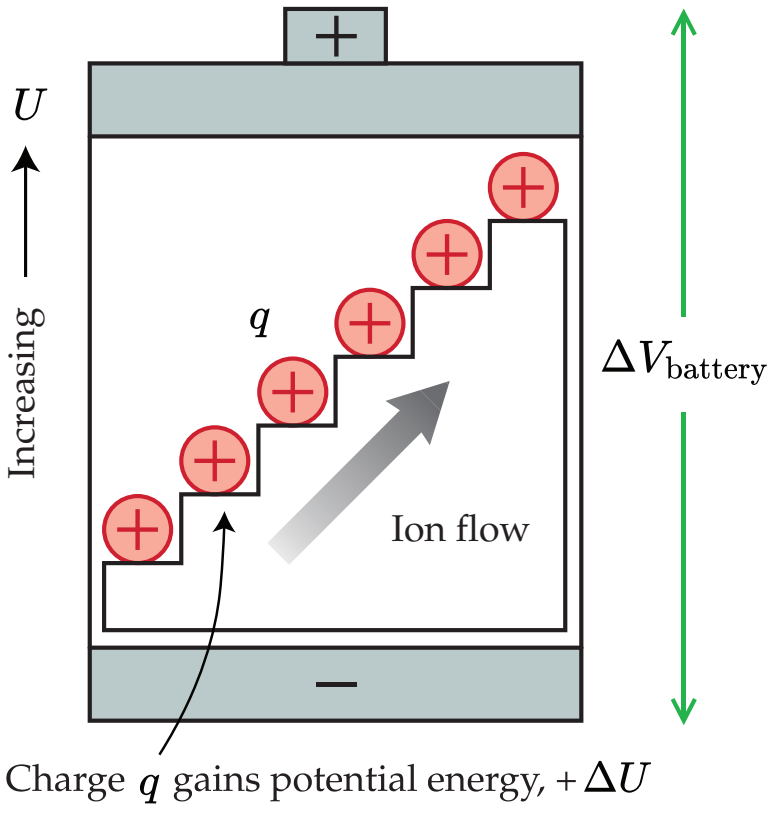
Everything you need to know about batteries in this course!

- An electric potential difference is created by separating positive and negative charges.
- A **Battery** does that by using *chemical reactions* to separate charges. The process is called *Electrolysis*.
- It pulls positive and negative charges apart, creating a potential difference between the terminals of the battery.
- When the chemicals are used up, the reactions cease, and the battery is *dead*.

Charge Escalator model of a Battery and its EMF

You must work on the system, hence the input power!

$$\Delta V = V_{pos} - V_{neg} = -\int_{+}^{-} E_r dr$$



Charge Escalator model of a Battery and its EMF
Whether to's and why for's!

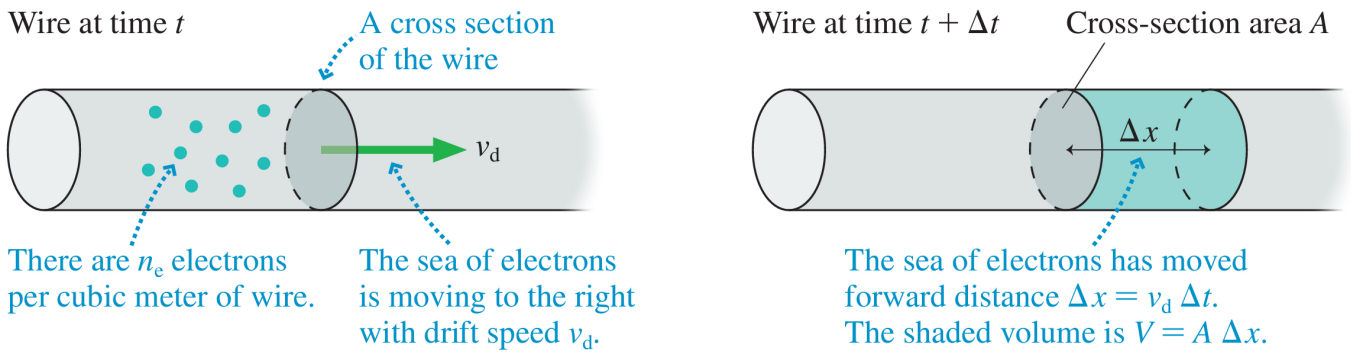
- The charge escalator lifts positive charges from the negative terminal to the positive terminal. This requires *work*, with the energy being supplied by the chemical reactions.
- The work done per charge is called the Electromotive Force (EMF) of the battery: $\mathcal{E} = \frac{W_{\text{chem}}}{q}$.
- The charge separation creates a potential difference $\Delta V_{\text{battery}}$ between the terminals. An ideal battery has $\Delta V_{\text{battery}} = \mathcal{E}$. It is measured in Volt [V] unit.
- **Limitations**: $\Delta V_{\text{battery}} < \mathcal{E}$ if current flows through the battery. The difference is usually small, and a battery can be considered ideal for simplicity.

What are Charge Carriers and Currents?

BRAC UNIVERSITY
Inspiring Excellence

Drift Velocity

How drift motion leads to current



The number of electrons that pass through the cross section during the time interval Δt is

$$\begin{aligned} \Delta Q &= I\Delta t \\ n_e V = n_e e(A\Delta x) &= I\Delta t \qquad [A\Delta x = V] \\ n_e eA(v_d \Delta t) &= I\Delta t \qquad [v_d \Delta t = \Delta x] \\ \therefore I &= n_e eAv_d \end{aligned}$$

Theory of Metallic Conduction

Where does this drift velocity even come from?

Suppose an electron just had a collision with an ion and has rebounded with velocity $\vec{v}_0$. The acceleration of the electron between collisions is

$$a = \frac{e}{m_e} E$$

E causes the x-component of the electron's velocity to increase linearly with time:

$$v = v_0 + a\Delta t = v_0 + \left(\frac{eE}{m_e}\right)\Delta t$$

The electron speeds up, with increasing K, until its next collision with an ion. A sea of electrons does the same.

A New Look at Drift Velocity

Why is it so small in magnitude?

The average speed at which the electrons are pushed along by the electric field is

$$v_d = \bar{v} = \left(\frac{e\tau}{m_e}\right) E,$$

where τ is the average value of Δt is the mean time between two successive collisions.

Define Electric Current

We are jumping a bit! I hope you don't mind!

An **electric current** is the net (positive) charge flowing per unit time, through a cross-section of a conducting wire.

Defining Electric Current in terms of Charge flowing, Drift Velocity:

$$i_e = n_e A v_d$$

$$I = e i_e = n_e e A v_d = \left(\frac{n e^2 A \tau}{m_e} \right) E.$$

Long Story Short

Once and for all

Electrons don't just magically move through a wire as a **current**. They move because an electric field inside the wire—a field created by a nonuniform surface charge density on the wire—pushes on the sea of electrons to create the **electron current**. It is measured in A C^{-1} unit or simply electrons s^{-1} unit.

Currents do not come out of the battery nor get inside it on the other end of the circuit. Even though electrons move relatively slowly (a drift speed of millimeters per second in a wire), the *effect* of current propagates near the speed of light because the electric field pushes all the electrons at once, like a chain reaction.

Incepting Ideas (1)

Hint: Know the difference between Electron and Electric current.

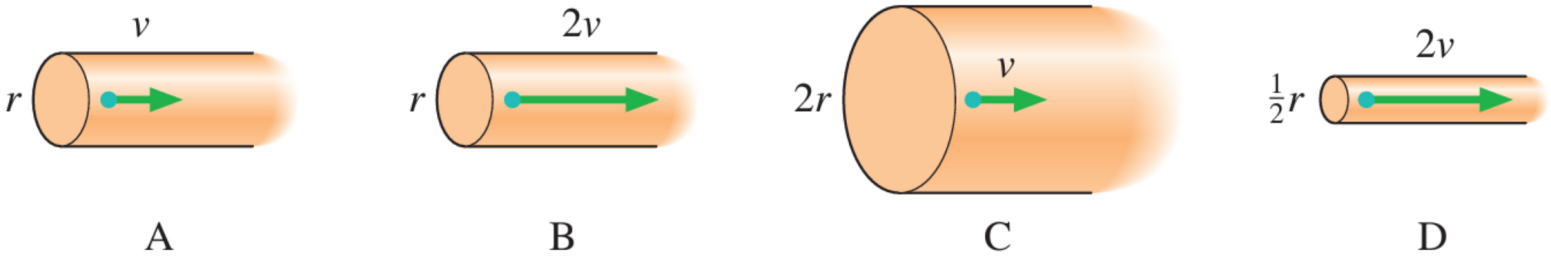
Q1: What are the electron and electric current in a 2.0 mm-diameter copper wire if the electron drift speed is 10 mm s^{-1}? The electron density for copper as 8.5×10^{28} m^{-3}.

Q2: A silver wire 2.6 mm in diameter transfers a charge of 420 C in 80 min. Silver contains 5.8×10^{28} free electrons per cubic meter. (a) What is the current in the wire? (b) What is the magnitude of the drift velocity of the electrons in the wire?

Incepting Ideas (2)

Hint: Faster drift speed makes faster currents

Q: These four wires are made of the same metal. Rank in order, from largest to smallest, the electric currents I_A to I_D.



Incepting Ideas (3)

Hint: Practice! Practice! Practice!

Some Problems to Practice at Home on Electric Current

Q 25.2 to 25.8, p-865, **Example** 25.1, p-844, 25.2, p-849 | Young-Freedman

Exercise Problem 25.14, 25.15 p-846 | YF

Checkpoint 3, p-755, **Sample Problem** 26.01, p-747, 26.03, p-752, 26.06, p-756 | Resnick-Halliday

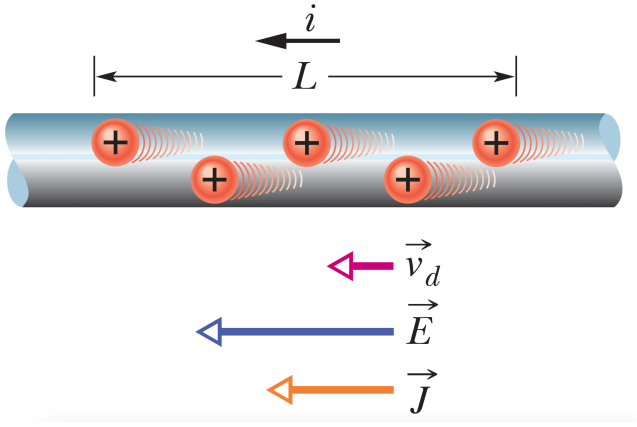
Exercise Problem 2, p-765, 8, 10, 19, 20, 23, p-709 | RH

The Current Density in a Wire

It helps you find the direction of the actual current

Define **Electric Current Density**:

$$J = \frac{I}{A} = n_e e v_d \ [\mathrm{A\,m^{-2}}]$$



It is a **vector**. It is used to measure current from a distribution where each section of the conductor may not have the same current through it.

$$I = \int \vec{J} \cdot d\vec{A} = J dA \cos\theta$$

Incepting Ideas (4)

Hint: Know when to use the scalar part of $\vec{J}$

Q1: A 1.0 A current passes through a 1.0 mm-diameter copper wire. What are the current density and the drift speed of the electrons in the wire? The electron density for copper as $8.5 \times 10^{28}\ \mathrm{m}^{-3}$.

Q2: The current density in a cylindrical wire of radius $R = 2.0$ mm is uniform across a cross-section of the wire and is $J = 2.0 \times 10^5\ \mathrm{A\,m}^{-2}$. What is the current through the outer portion of the wire between radial distances $\frac{R}{2}$ and R?

Enter Ohm's Law

BRAC UNIVERSITY
Inspiring Excellence

Ohm's Law in the making: Conductivity and Resistivity

Not what you expected, eh?

Relating Current Density with Electric Field Intensity

$$J = \frac{I}{A} = \left(\frac{n_e e^2 \tau}{m_e}\right) E$$

$$J = \sigma E$$

$$E = \rho J; \quad \therefore \sigma = \frac{1}{\rho}$$

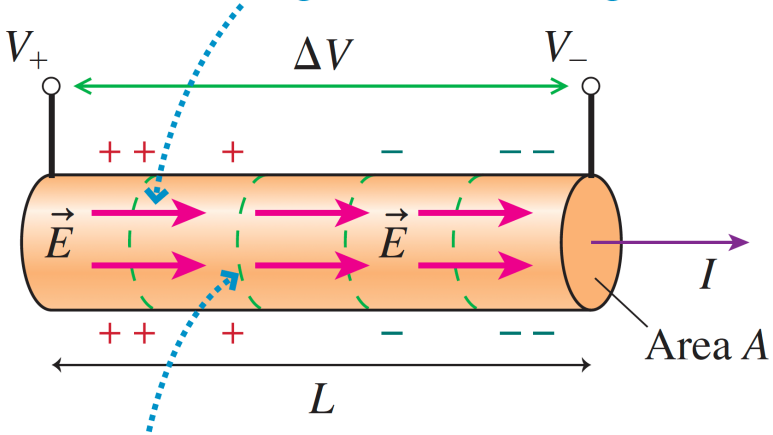
σ is measured in $\Omega^{-1}\,\mathrm{m}^{-1}$,

ρ is resistivity; measured in $\Omega\,\mathrm{m}$.

What is Resistance?

Obstruction to current

The potential difference creates an electric field inside the conductor and causes charges to flow through it.



Equipotential surfaces are perpendicular to the electric field.

Ohm's Law

Ahh! The familiar form!

Ohm's Law in 3D, $E = \frac{\Delta V}{L} = \rho \frac{I}{A} \Rightarrow \Delta V = \left(\frac{\rho L}{A}\right) I$

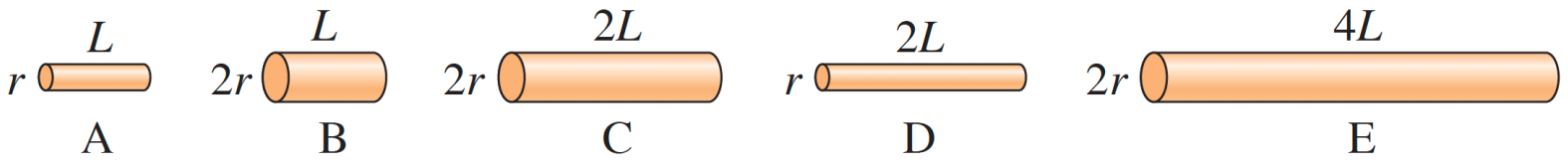
with Resistance, $R = \frac{\rho L}{A}$; measured in Ω

Ohm's Law (Experimental Form) $I = \frac{\Delta V}{R}$

Incepting Ideas (5)

Hint: Same material means same ρ and σ

Q1: The wires are all made of the same material. Rank in order, from largest to smallest, the resistances R_A to R_E of these wires. Explain.



Q2: The wires are all made of the same material. All wire segments are held at equal potential differences. Rank in order, from largest to smallest, the currents I_A to I_E of these wires. Explain.

What does Ohm's Law Say about Circuits?

Its physical significance and practical usage

- Current is caused by an electric field exerting forces on the charge carriers
- The current density, and hence the current $I = JA$, depends linearly on the strength of the electric field. To double the current, you must double the strength of the electric field that pushes the charges along
- The current density also depends on the *conductivity* of the material. Different conducting materials have different σ because they have different values of the electron density n_e and, especially, different values of the mean time τ between electron collisions with the lattice of atoms.

Incepting Ideas (6)

Hint: Use the experimental form of Ohm's law

Q: A 2.0 mm-diameter Aluminum wire carries a current of 800 mA. The conductivity of the Aluminum wire is $3.5 \times 10^7\ \Omega^{-1}\ \mathrm{m}^{-1}$.

- What is the electric field strength inside the wire?
- What is the resistivity of the wire?
- Find the resistance of 1 m section of that same wire.

Homework Practice Problem: Try it Yourself

Q: The free-electron density in a copper wire is $8.5 \times 10^{28}\ \mathrm{m}^{-3}$. The electric field in the wire is $0.0600\ \mathrm{V\,m}^{-1}$. (a) What is the potential difference between two points in the wire that are separated by 20.0 cm? (b) What is the drift speed of the electrons in the wire?

Incepting Ideas (7)

Hint: Use the 3D form of Ohm's law

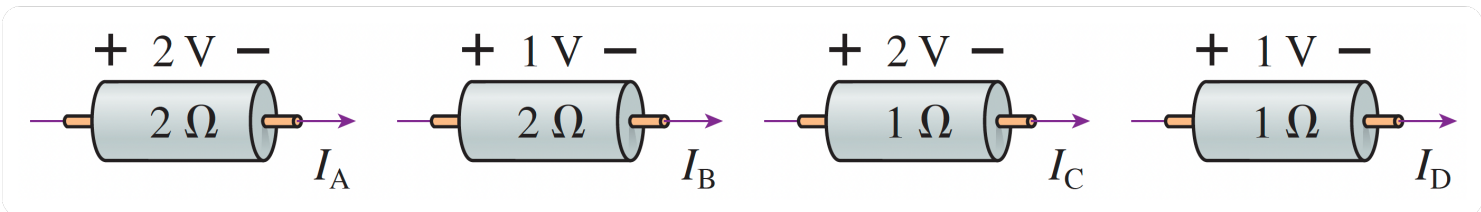
Q: A 15 cm-long Nichrome wire is connected across the terminals of a 1.5 V battery. The conductivity of the Nichrome wire is $6.7 \times 10^5\ \Omega^{-1}\,\mathrm{m}^{-1}$.

- What is the electric field inside the wire?
- What is the current density inside the wire?
- If the current in the wire is 2.0 A, what is the wire's diameter?

Incepting Ideas (8)

Hint: Use the experimental Ohm's law

Q: Rank in order, from largest to smallest, the currents I_A to I_D through the four resistors.



Incepting Ideas (9)

Hint: Practice! Practice! Practice!

Some Problems to Practice at Home on Ohm's Law

Q 25.2 to 25.8, 25.12, p-865, **Example** 25.14, 25.15, p-866, 25.20, 25.21, p-867 | Young-Freedman

Exercise Problem 25.14, 25.15 p-846 | YF

Checkpoint 3, p-755, 4, p-757 **Sample Problem** 26.04, p-747, 26.06, p-756 | Resnick-Halliday

Exercise Problem 5, 6 p-765, 19, 20, 23, 25, p-766, 29, 31, 34, p-767, 58, 69, p-769 | RH

That is it for today!

BRAC
Inspiring Excellence